BIE 599: Phase 1 project
Healthcare Chatbot

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Project Guide:
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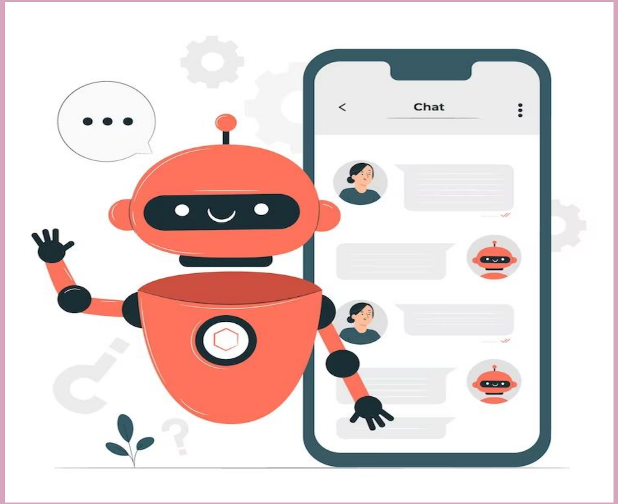
Introduction

chatbot is a computer program that simulates and processes human conversation allowing humans to interact with digital devices



https://siamcomputing.com/digital-transformation/chatbot/

Chatbot in Healthcare

clinic

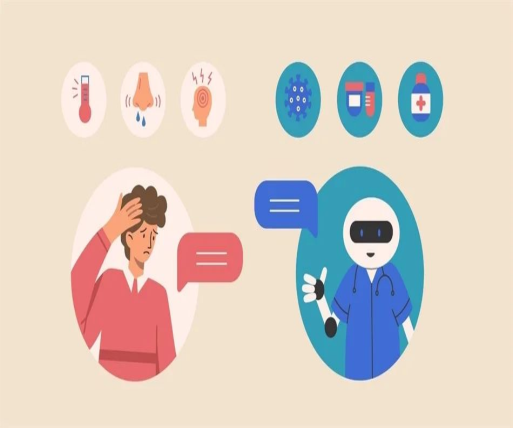
Old method

Doctor Consultation

New method

Home

AI consultation

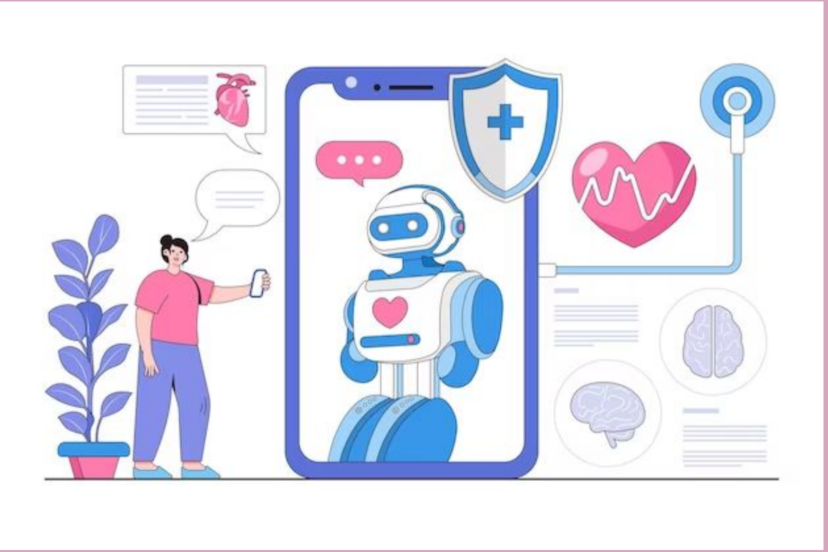


24/7 Accessibility
Rapid response
Scalability
Cost effective
Convenience
Early detection

https://www.freepik.com/free-photos-vectors/doctor-patient

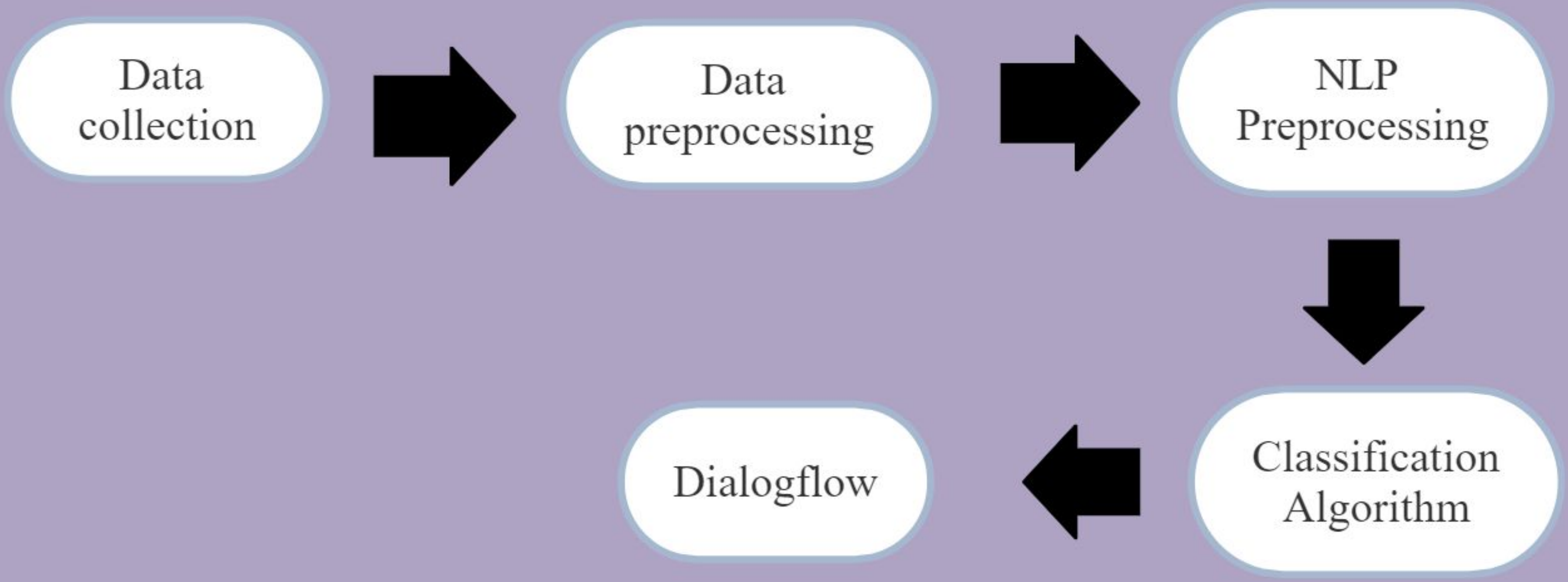
Objective

To create healthcare chatbot using machine learning

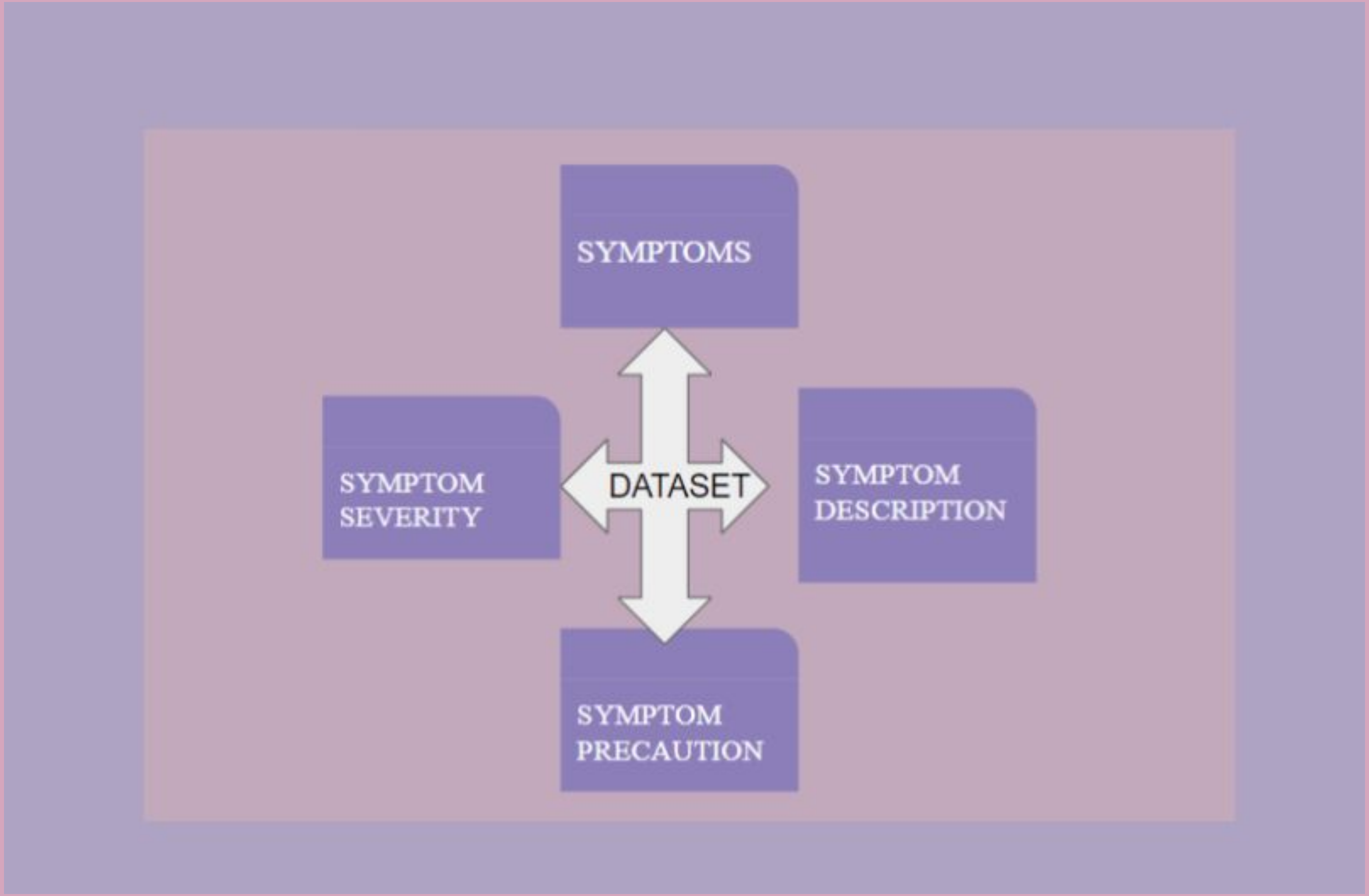


https://www.freepik.com/free-photos-vectors/doctor-patient

Methodology



Data collection/Dataset



Data Preprocessing

Counter vectorizer Technique : text to matrix of individual word/token

*	Symptom 1	Symptom 2
Disease 1	0	1
Disease 2	0	0

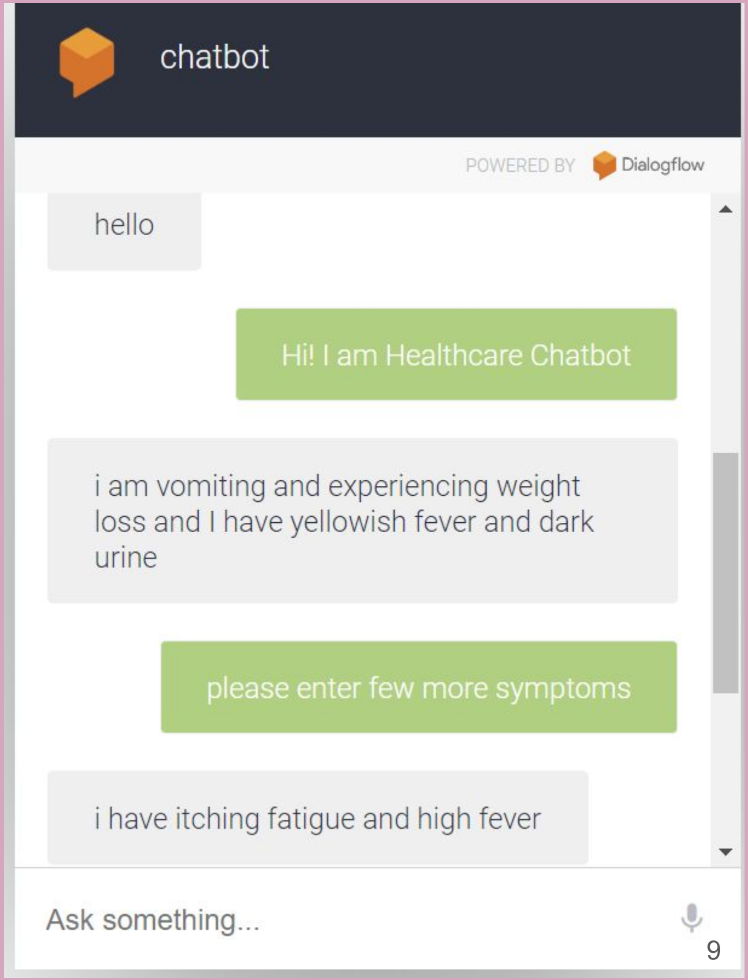
NLP Preprocessing

- Tokenization
- Stemming
- TF-IDF

Software

It is a cloud-based natural language processing (NLP) platform developed by Google.

https://dialogflow.cloud.google.com/



Result will be in the form of
Disease description
Disease precaution and its severity

Literature Survey

S.NO	JOURNAL/YEAR	AUTHORS	TITLE	METHODS	COMMENTS
1	International Conference for Emerging Technology (INCET) 2021	Sagar Badlani, Tanvi Aditya,Meet Dave ,Sheetal Chaudhari	Multilingual Healthcare Chatbot Using Machine Learning	Machine learning algorithms(SVM), NLP	Pre processing issue
2	International Conference on Computer and Information Technology (ICCIT) 2019	Moshiur Rahman et al	An Implementation of Machine Learning Based Bangla Healthcare Chatbot	Machine learning(Random forest,KNN), cosine similarity	SVM - best classifier, accuracy problem

3	Journal of Emerging Technologies and Innovative Research (JETIR) 2022	Mohammed Juned, Farhat Dalvi	AI Healthcare Chatbot	Random forest,d Flask Framework	Classifier model

Timeline

Month	Work to be done
Aug-sep	Data collection and data preprocessing
oct	NLP preprocessing
Nov	Dialogflow

References

- M. M. Rahman, R. Amin, M. N. Khan Liton and N. Hossain, "Disha: An Implementation of Machine Learning Based Bangla Healthcare Chatbot," 2019 *22nd International Conference on Computer and Information Technology (ICCIT)*, Dhaka, Bangladesh, 2019, pp. 1-6, doi: 10.1109/ICCIT48885.2019.9038579.
- R. B. Mathew, S. Varghese, S. E. Joy and S. S. Alex, "Chatbot for Disease Prediction and Treatment Recommendation using Machine Learning," *2019 3rd International Conference on Trends in Electronics and Informatics (ICOEI),* Tirunelveli, India, 2019, pp. 851-856, doi: 10.1109/ICOEI.2019.8862707.
- P. Srivastava and N. Singh, "Automatized Medical Chatbot (Medibot)," *2020 International Conference on Power Electronics & IoT Applications in Renewable Energy and its Control (PARC)*, Mathura, India, 2020, pp. 351-354, doi: 10.1109/PARC49193.2020.236624.